

Calculate the divergence of the following vector functions:

(a) $\mathbf{v}_a = x^2\hat{\mathbf{x}} + 3xz^2\hat{\mathbf{y}} - 2xz\hat{\mathbf{z}}$

(b) $\mathbf{v}_b = xy\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + 3xz\hat{\mathbf{z}}$

(c) $\mathbf{v}_c = y^2\hat{\mathbf{x}} + (2xy + z^2)\hat{\mathbf{y}} + 2yz\hat{\mathbf{z}}$

Solution:

The divergence of a vector $\mathbf{v}$ is defined as

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

(a) $\nabla \cdot \mathbf{v}_a = 2x - 2x = 0$

(b) $\nabla \cdot \mathbf{v}_b = y + 2z + 3x$

(c) $\nabla \cdot \mathbf{v}_c = 2x + 2y$

□
